A Machine Learning-Based Groundwater Level Predictor for Sustainable Water Management in Karnataka

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Abstract -The proposal of this project outlines the development of a machine-learning-based Groundwater Level Predictor on behalf of Karnataka, with the focus on water sustainability. The project goal is to reduce the acute pressure on the groundwater resources caused by increased demand and climate change through offering predictive mechanisms of proactive stewardship. The objective is to create a powerful model that can lead to the correct prediction of groundwater level. To do so, a combined spatio-temporal database has been built out of varying sources, state-of-the-art machine-learning algorithms, which are mainly the Random Forest, are generated, evaluated, and validated, and actionable information regarding the main contributors to the variability of groundwater levels is derived. The deliverables include a validated predictive model, a full dataset, and a technical report, which will be used to assist the government agencies in allocating data-based water and prepare for drought, and evaluate their policies to make Karnataka water-secure. To address the growing water demand and the effects of climate change, the initiative aims to provide administrative instruments for stewardship. The main objective is the formulation of a viable model that predicts monthly groundwater levels at the taluk level, using a spatio-temporal dataset that is obtained through different sources. The state-of-the-art machine-learning models, with the top suggestion being the Random Forest, will be implemented and evaluated against high-performance validation. Above prognostication, the project will measure and determine important parameters that drive changes in the groundwater level, hence generating actionable knowledge. The deliverables will include a validated prediction model, a complete dataset, and a technical report presenting results. These will initiate the delivery of data-driven decision support to the government agencies related to water allocations and drought preparedness and policy-making, which will form part of the future of water in Karnataka and its sustainability over the long term.

INDEX TERMS- Machine Learning, Groundwater Level Prediction, Random forest, sustainable water management, Karnataka, Spatio-temporal dataset, streamlit, time series cross-validation, water allocation, drought preparedness, policy assessment.

I. INTRODUCTION

Groundwater is a central natural resource that is utilized in farming, industrial, and household consumption. In India, it supplies over 60 percent of the irrigation water and around 80 percent of the drinking water. Over-extraction, urbanization, and climate change are the main factors that lead to its rapid depletion, affecting the quality and quantity of available groundwater. Even though organizations like the Central Ground Water Board (CGWB) [1] are observing the water levels in almost 26,000 wells and tend to rely more on Digital Water Level Recorders (DWLRs), these are not predictive, as they are aimed at data gathering and transmission. Traditional modelling methods, such as the Autoregressive Integrated Moving Average (ARIMA), do not adequately represent the nature of highly non-linear multi-factor interaction of aquifer systems [3].

In order to seal this gap, a computer program called Ground Water Level Predictor has been created. The application combines the meteorological, hydrological, geospatial, and demographic factors in one platform. It uses a Random Forest regression model in Python and uses the Streamlit deployment platform to predict the level of groundwater in its spatial and temporal context. The interactive dashboard gives maps, time-series visualisations, and soil-water depth profiles, and therefore enables stakeholders to understand and plan groundwater utilisation.

The CGWB has a comprehensive system with about 26000 wells of observation and continues to install digital recorders to improve information collection. Although their use is effective in collecting and transferring information, these systems do not often include powerful predictive analytics. Conventional hydrological models like ARIMA have been shown to lack the ability to capture the intricate non-linear interdependencies that can be found in the context of aquifer systems, and this thus restricts their effectiveness in forecasting.

In recognition of this rather deficient feature, the Ground Water Level Predictor platform has been developed. This complex system alleviates the existing deficiencies by converting an extensive combination of meteorological, hydrological, geospatial, and demographic indicators into a unitary, easily accessible system. Based on a powerful Random Forest regression model, built in Python, the system is executed through Streamlit; therefore, it allows exact forecasting of groundwater levels both spatially and temporally.

The Groundwater Level Predictor presents an interactive dashboard to enable the stakeholders with an actionable understanding. The dashboard contains dynamic maps of the groundwater distribution, time series illustrating past and future water-level changes, and profiles of the soil-water depth to gain a better picture of the groundwater process and, therefore, base strategic decisions and sustainable management of groundwater.

II. EXISTING METHODS

A. Current Groundwater Monitoring Practices

The government, in its enforcement of groundwater monitoring, through the Central Ground Water Board (CGWB)[1] and Karnataka Groundwater Authority (KGWA), is aided by an elaborate system of observation wells across the state[4]. These networks allow for documentation of long-term aquifer condition but provide no predictive capabilities. Decisions are generally made by extrapolating from trends or through some other subjective means, which is not conducive to water management in a proactive manner.

B. Traditional Statistical Models

The initial endeavors for groundwater level predictions relied on time-series models, including ARIMA, linear regression, and moving averages[6]. These models rely on linearity and stationarity assumptions, which are unreasonable, considering the non-linear and multi-factorial character of aquifer systems subject to rainfall variability, soil heterogeneity, and the pressure associated with human populations. Moreover, they are only capable of prediction within a heterogeneous hydrogeological context, like Karnataka.

C. Machine Learning and Deep Learning Approaches

In the last few years, machine learning has been used to address these problems. Tree-based ensemble models, such as Random Forest and Gradient Boosting, can manage mixed numerical and categorical inputs (rainfall, soil type, population, spatial coordinates) and related models consistently outperform classical statistical models [7]. Deep learning approaches, such as Long Short-Term Memory (LSTM) networks, as well as hybrid CNN-LSTM architectures, have been utilized to capture temporal dependencies and spatial attributes in groundwater datasets.

D. Gap in User-Friendly Deployment

Even though they hold promise, there are still many machine learning and deep learning models for groundwater prediction that are limited to research prototypes and have not yet been deployed in a format that could directly assist with planning and policy in a public-facing interactive manner.

E. Need for an Integrated Solution

Currently, Karnataka lacks a cohesive system that includes excellent groundwater datasets, cutting-edge machine learning models, and an interactive dashboard. This project is motivated by this gap, using an interpretable Random Forest regression model, which is implemented using a Streamlit-supported web application. This increases predictive accuracy and also a certain degree of transparency in forecasting among the planners, researchers, and the local communities.

III. PROPOSED METHODOLOGY

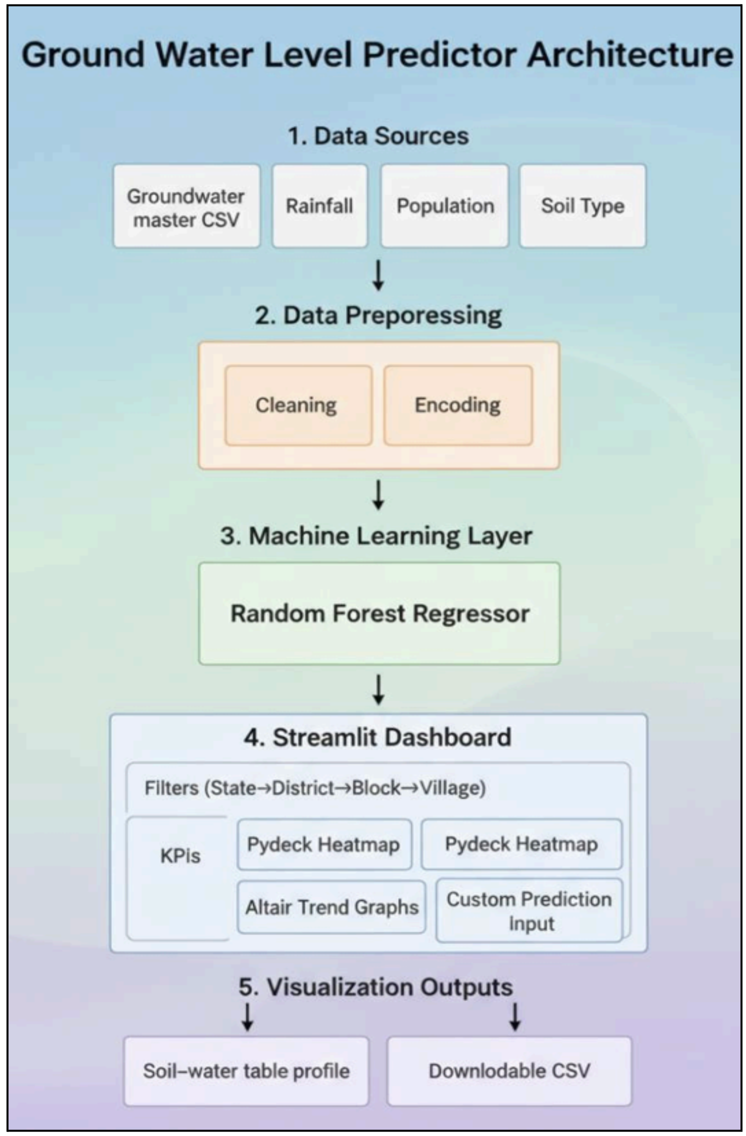

Fig. 1 System Architecture

A. Data Collection

This study employs a consolidated dataset assembled from official sources such as the Central Ground Water Board (CGWB)[1] and the Karnataka Ground Water Authority (KGWA). This data has spatial coordinates (latitude, longitude), observed groundwater levels (WL in meters below ground level), annual rainfall, block population, and predominant soil type for each monitoring location.

B. Data Preprocessing

Before modeling, the raw data is cleaned in a formal cleaning process in Python (pandas):

1. Renaming and standardizing field names
   – (LATITUDE → lat, LONGITUDE → lon).
2. Converting key fields (lat, lon, WL, rainfall, population) to numeric values.
3. Parsing and validating dates.
4. Dropping rows with missing critical values
5. Encoding the categorical variable Predominant_Soil_Type as one-hot encoding for machine learning input.

In this way, the model has consistent, analysis-ready data as inputs.

C. Model Training – Random Forest Regression

The prediction engine of the system is a Random Forest Regressor implemented with scikit-learn. The feature vector consists of rainfall, block population, predominant soil type (encoded), latitude, and longitude. The target variable is the observed groundwater level (WL in mbgl).

- Algorithm: RandomForestRegressor with 100 estimators, random_state=42.
- Advantages: Handles mixed numerical and categorical data, captures non-linear relationships, and is robust to outliers.
- Caching: The trained model is cached with @st.cache_resource to speed up app performance.

D. Dashboard Development – Streamlit Interface:

We built an interactive dashboard using Streamlit to share the outputs of the model in the following way:

I. Hierarchical Filters -Users can drill down from State → District → Block → Village to access data for each administrative unit.
II. Maps - Pydeck heatmaps (for groundwater variation) and scatterplots (showing observation locations) depict locations.
III. Counts of Water Level, Rainfall, Soil Type, and Population - displays matched to scales appear in real time.

Prediction - Users can provide soil type, rainfall, population, latitude, and longitude to estimate a possible water level.

E. Soil and Water Table Visualization:

A bespoke HTML element displays a color-coded soil chart with a living water table marker, which can easily help the user without technical expertise to understand the estimated depth of the groundwater footing and the strata of the soils.

F. Validation and Performance Metrics:

To measure the prediction reliability, the model was measured with the help of traditional regression measures, the root mean squared error (RMSE), the mean absolute error (MAE), and the coefficient of determination (R^2). These measures are used to determine the predictive accuracy and guide the next refinement of the model.

G. Deployment and Scalability:

It is a completely web-based system so that the planners, researchers, and the local communities can use it without having to download specialized software and install it. Streamlit, being a lightweight architecture, allows smooth updates when new data is available, and any future modeling improvements (e.g, LSTM or CNN-LSTM architecture to make long-term predictions) can easily be incorporated.

IV. OBJECTIVES

The main objective of the project is to design and deploy a strict data-driven system, which can predict and visualize the groundwater level in the entire state of Karnataka using machine learning approaches. The targeted goals include the following:

A. *Data Integration:*
The task will be to gather, synthesize, and purify groundwater, rain, soil category, demography, and geospatial data from believable sources to form a comprehensive and analysis-ready dataset.

B. *Model Development:*
To implement and train a Random Forest regression model that can reliably predict groundwater level (in meters below ground level), rainfall, soil type, population, and spatial coordinates as inputs.

C. *Interactive Dashboard:*
To build a user-friendly Streamlit dashboard that allows end users to filter data by State, District, Block, and Village while viewing actual vs predicted groundwater level.

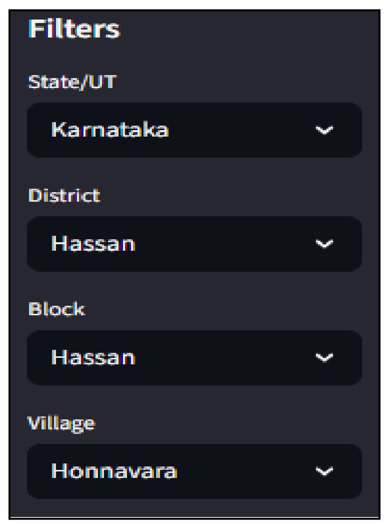

Fig. 2 Filter Interaction

D. *Visualization and Interpretation:*
To create user-friendly visualizations, Pydeck heatmaps, Altair time-series charts, and a custom soil–water table profile, that result in easily interpretable predictions and underlying data.

E. *Performance Validation:*
To assess the model using standard regression metrics (RMSE, MAE, R_2) and to verify that the predictions are appropriate for making policy and planning decisions.

F. *Scalability and Future Extension:*
To offer a design that allows for the updating of the model with new data, and to extend the model to use more complex models, including LSTM or CNN-LSTM, for long-term forecasting

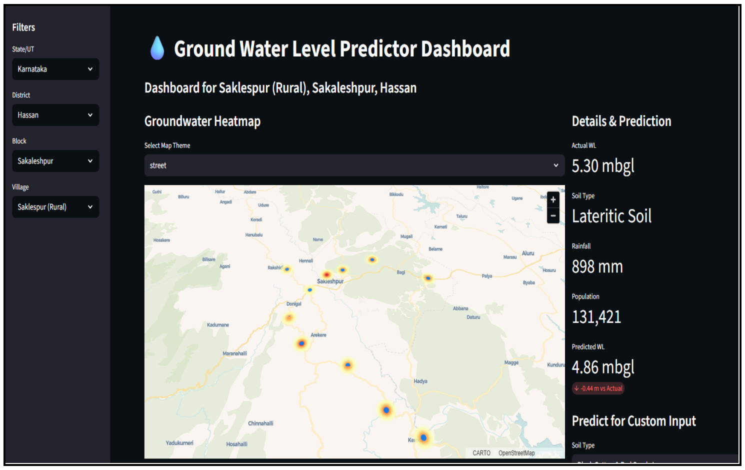

Fig. 3 Web Dashboard for Visualization

V. SYSTEM DESIGN AND IMPLEMENTATION

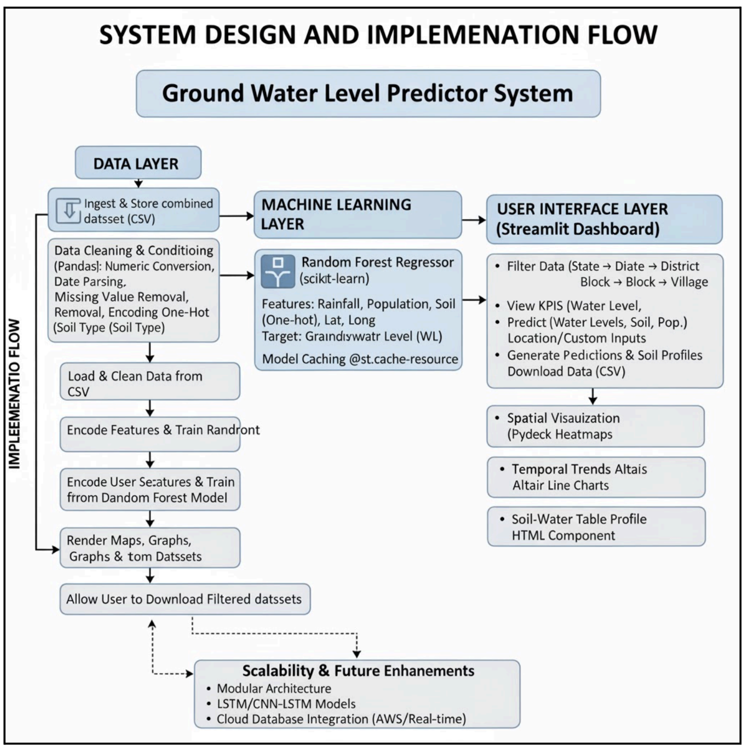

Fig.4 System Design

A. System Overview

The Groundwater Level Predictor system functions as a modular pipeline, converting the raw data of groundwater and environmental factors to predictions and to visualisations that can be acted upon, made up of 3 layers: The Data Layer, the Machine Learning Layer, and the User Interface Layer.

B. Data Layer

The data layer ingests and stores the combined dataset (groundwater_master_data.csv) for latitude, longitude, observed groundwater levels, yearly rainfall, block population, and the predominant soil type. The data was subsequently cleaned and conditioned for use in the Python (pandas) environment. The data conditioning included Numeric conversion, parsing dates, removal of missing values, and one-hot encoding of soil types.

C. Machine Learning Layer

The predictive engine is a Random Forest Regressor using scikit-learn.

- **Features:** Rainfall, Block Population, Predominant Soil Type (One-hot encoding), Latitude, Longitude.
- **Target:** Groundwater level (WL in metres below ground level).
- **Strong points:** The model elicits non-linear features and interactions of features, supports mixed-type features, and gives features of importance that are easily interpretable. The computational model is stored in memory, with the help of the decorator of the st. Cache resource, which means that retraining is not required with every user interaction.

D. User Interface Layer (Streamlit Dashboard)

The system has an interactive Streamlit dashboard as the front end that enables the stakeholders to:

1. **Filter Data:** Drill down from State → District → Block → Village.
2. **View KPIs::** Visualize real water level, rainfall, soil type, and population data of the chosen location.
3. **Predict Water Levels:** Develop predicted groundwater levels based on the corroborated village selected or custom inputs (soil type, rainfall, population, latitude, longitude).

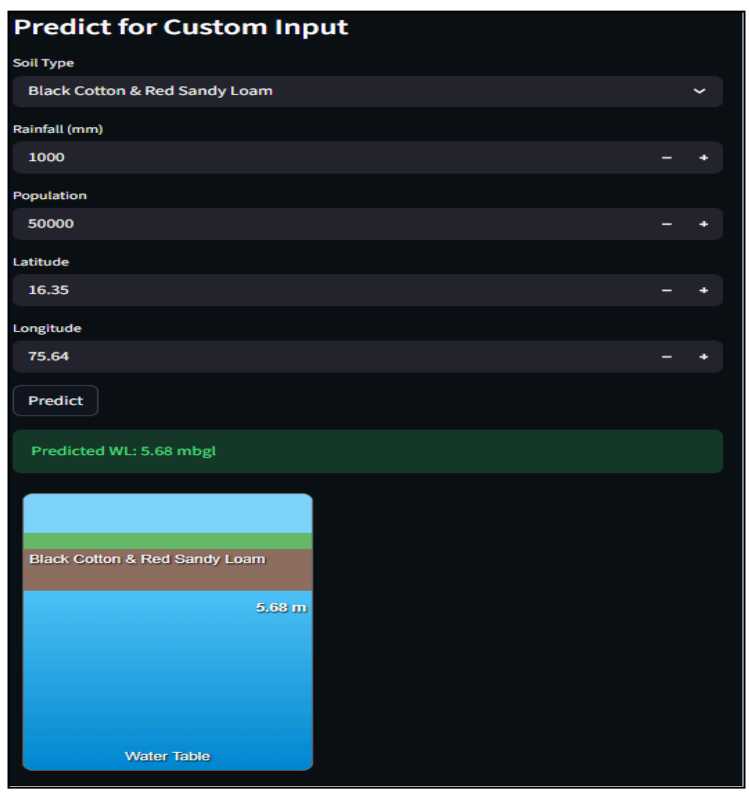

Fig. 5 Water Level Prediction

4. **Download Data:** Export data features can be used to create a CSV file of filtered data to be analyzed offline.

E. Visualization Components

There are two primary visualization modules that are used to improve interpretability:

- **Spatial Visualization:** GIS Pydeck heatmaps and scatter plot bars depict the spatial change in groundwater level and pinpoint the points of observation.

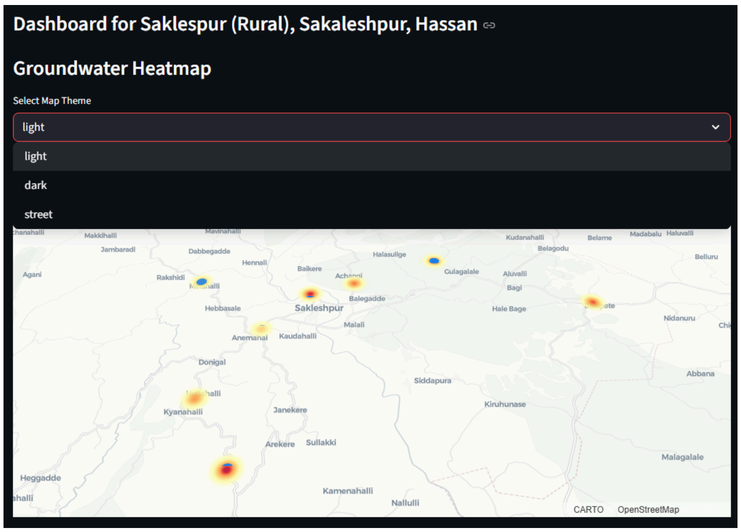

Fig. 6 HeatMap Visualization

- **Temporal Trends:** Altair line charts show trends of monthly or annual groundwater in the districts.

- **Soil–Water Table Profile:** A custom HTML component renders colour-coded soil layers with a dynamic water-table indicator based on predicted depth.

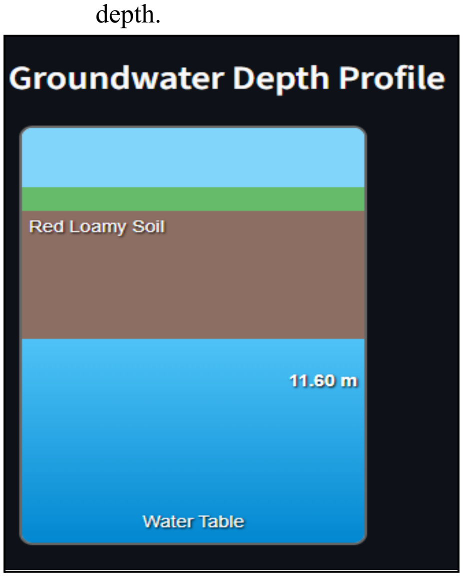

Fig. 7 Groundwater Depth Profile

F. Implementation Flow

The system workflow proceeds as follows:

1. Load and clean data from the CSV file.
2. Encode features and train the Random Forest model.
3. Accept user selections and/or custom inputs from the dashboard.
4. Generate predictions and KPIs.
5. Render maps, graphs, and soil profiles.
6. Allow the user to download filtered datasets.

G. Scalability and Future Enhancements

The use of modular architecture permits the system to be updated with new data and customized using leading-edge models like LSTM or CNN-LSTM for updated long-term forecasting, more geospatial layers built in, and one of the capabilities is also integrated with cloud databases like AWS or similar for real-time operation.

VI. CONCLUSION

Demonstrating water resource sustainability through efficient groundwater management, the Ground Water Level Predictor described in this paper shows how machine learning can be applied with tools such as interactive dashboards to understand and predict groundwater levels at a fine scale. The synthetic use of groundwater observations with rainfall, soil type, population, and geospatial coordinates shows how a Random Forest regression model can provide trustworthy predictions of groundwater levels at a fine scale.

A Stream lit-based dashboard provides all stakeholders with a simple and yet powerful interface for reviewing historical data, generating customized forecasts, and especially visualizing soil–water profiles and spatial patterns. This accessibility will close the gap between advanced analytics and ground-level decision-making, and thus provide the planners, researchers, and local communities with empirically-based water-management solutions.

The current project demonstrates that tree-based ensemble methods (Random Forest) provide acceptable benchmarks for groundwater prediction methods in Karnataka, outperforming traditional statistical models with the ability to capture complex non-linear relationships and municipality-level forecasts. The modular design would also allow additional data sources to be added or an upgrade to more sophisticated deep learning models (e.g., LSTM or CNN-LSTM) for long-term forecasting.

VII. ACKNOWLEDGMENT

The authors express their sincere gratitude to Presidency University, Bengaluru, for providing the academic infrastructure, support in implementing this project. The authors would also like to thank **Dr. Yamanappa**, Assistant Professor, School of Computer Science and Engineering, for his continuous guidance, encouraging suggestions, and motivation for authoring the Ground Water Level Predictor system.

We also appreciate the effort of our Department Head, **Dr. Pravinthraja S**, and Project Coordinator, **Dr. H. M. Manjula**, and School Coordinators, **Dr. Sampath A. K.,** and **Dr. Geetha A**., for their administrative and technical assistance.

Lastly, we thank the **Central Ground Water Board** (CGWB) [1] and the Karnataka **Ground Water Authority** (KGWA) for providing groundwater data in the public domain, which was used as the basis of our dataset, and the open-source developer community of Streamlit and scikit-learn for making the tools employed in this study.

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